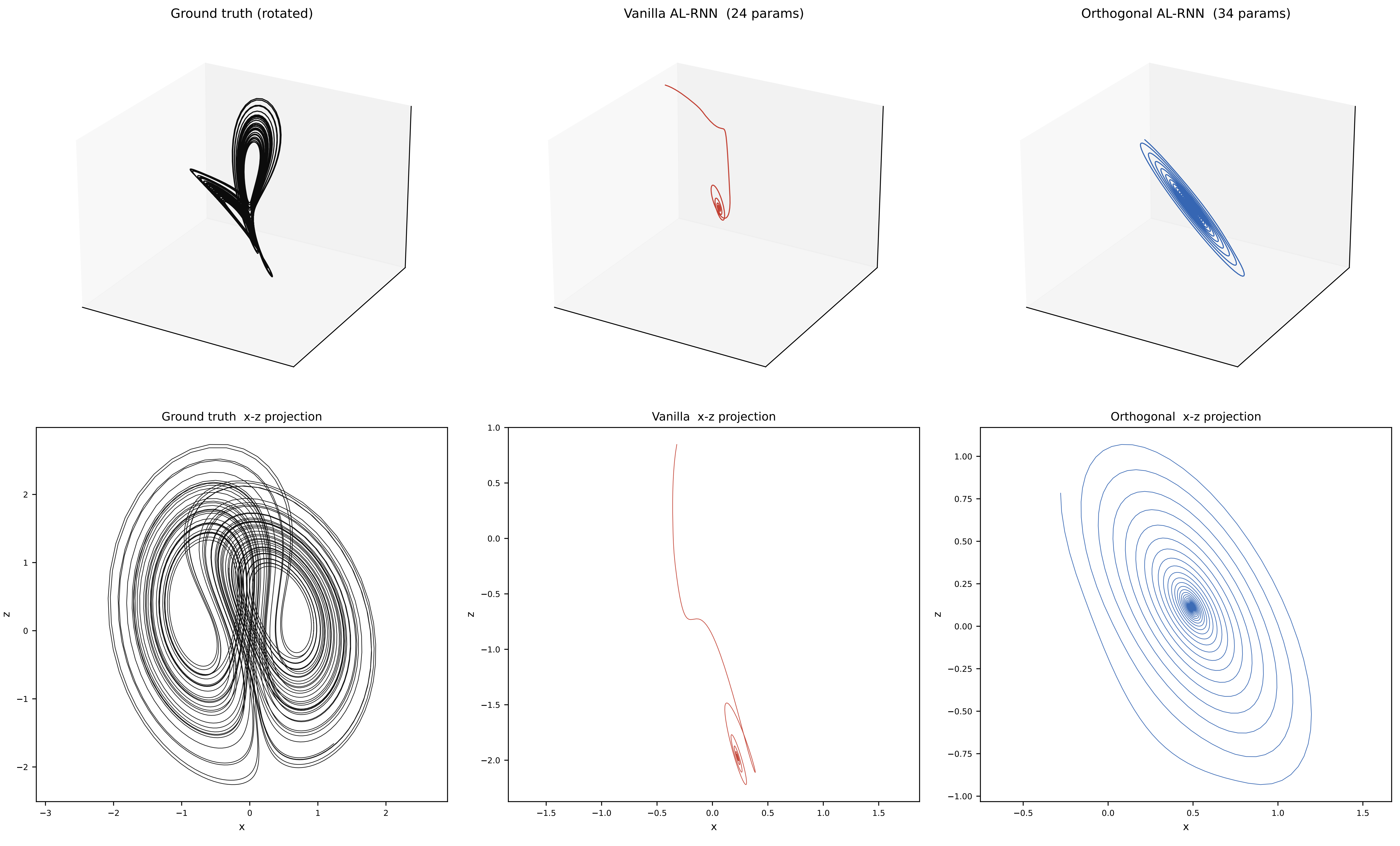
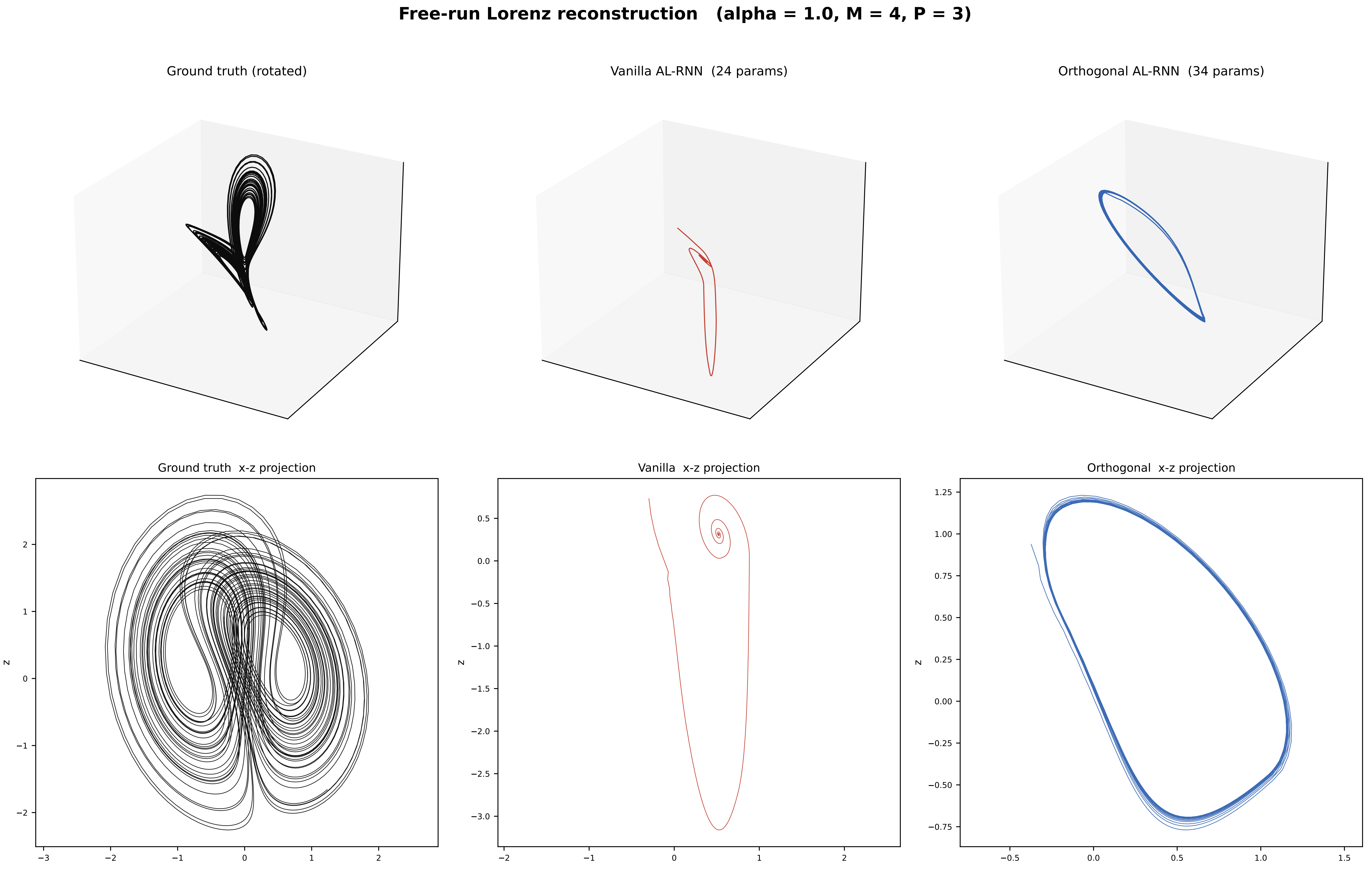
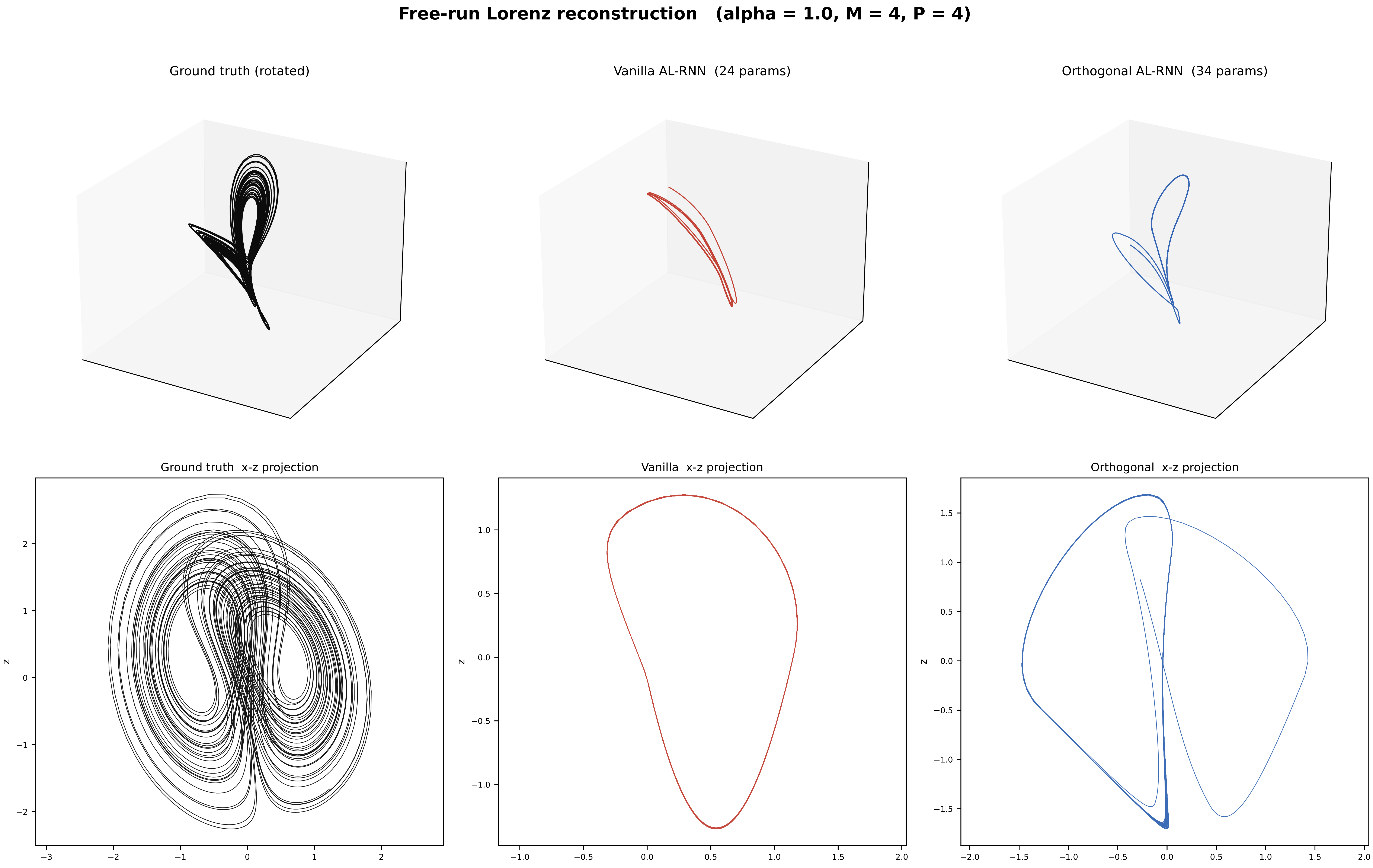
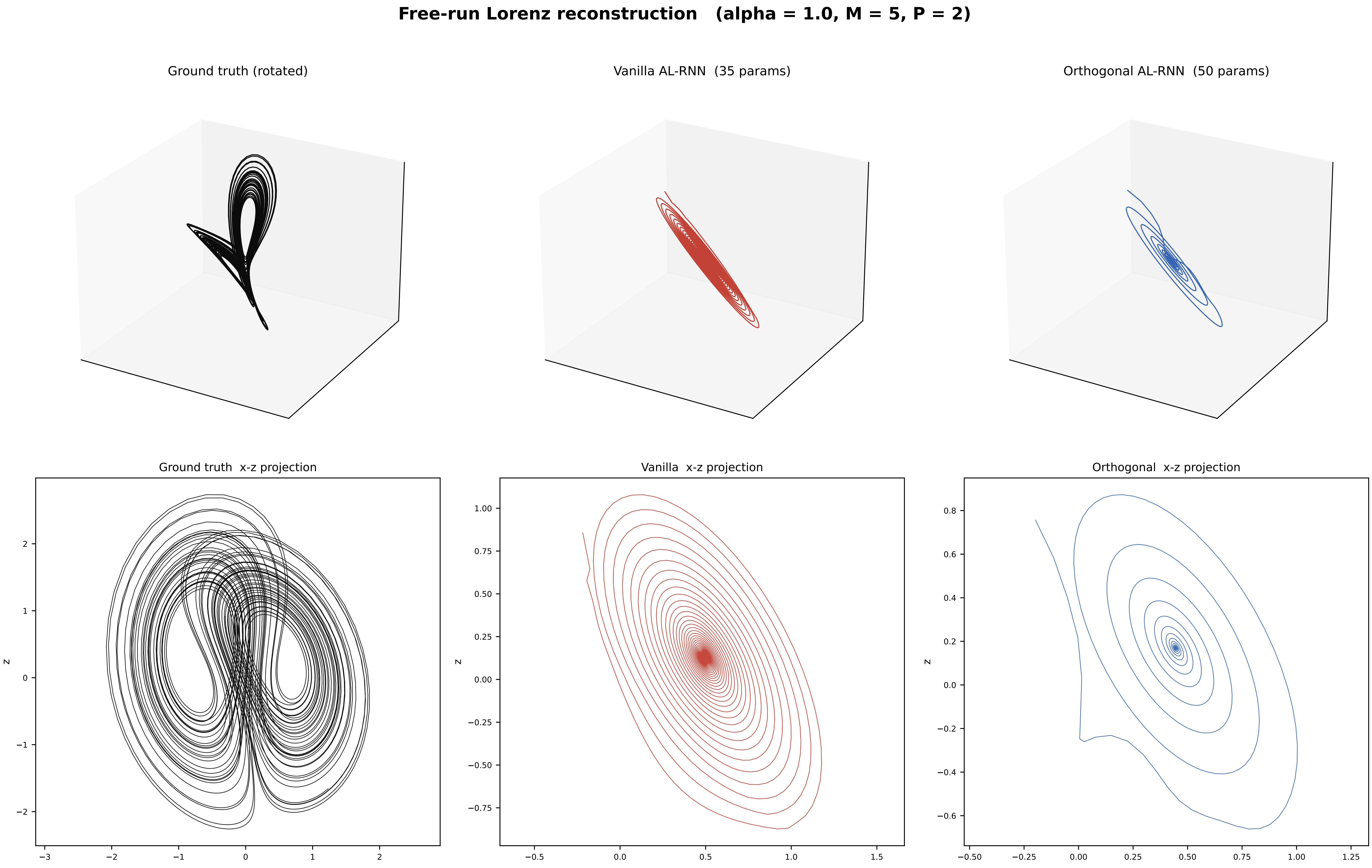
Free-run Lorenz reconstruction (alpha = 1.0, M = 4, P = 2)



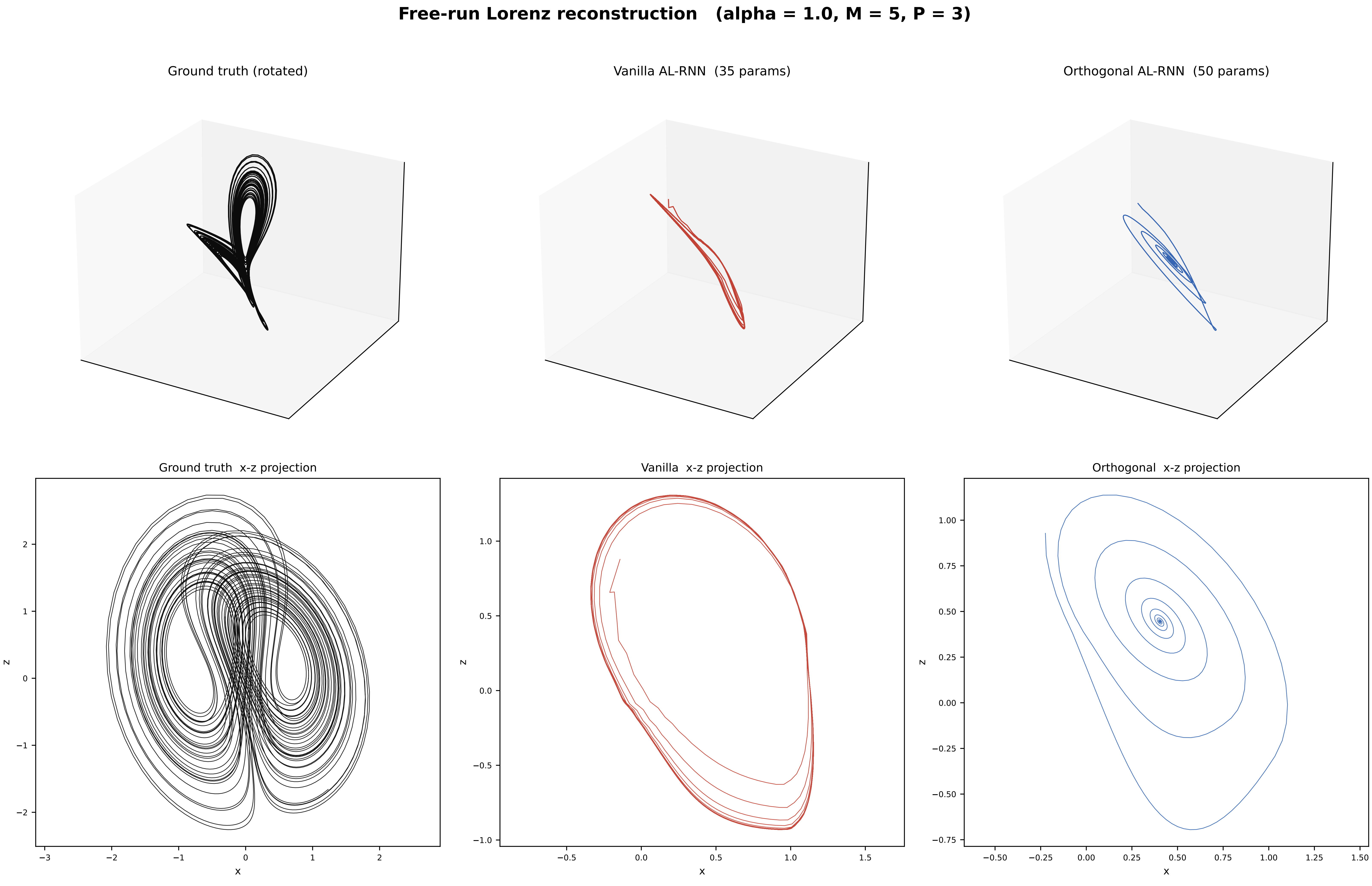


Free-run Lorenz reconstruction (alpha = 1.0, M = 4, P = 3)
Ground truth (rotated)
Vanilla AL-RNN (24 params)
Orthogonal AL-RNN (34 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z

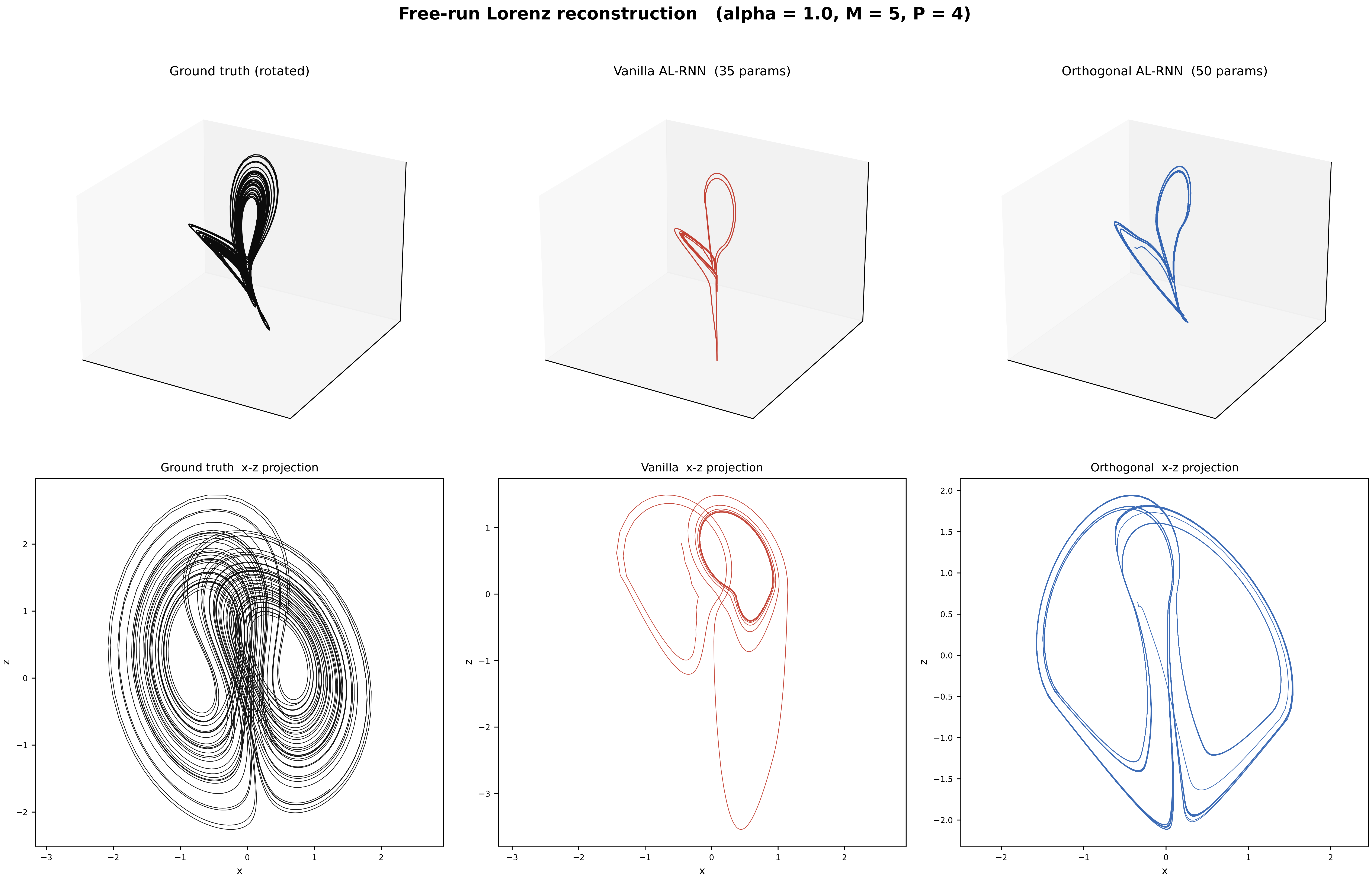

Free-run Lorenz reconstruction (alpha = 1.0, M = 4, P = 4)
Ground truth (rotated)
Vanilla AL-RNN (24 params)
Orthogonal AL-RNN (34 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z

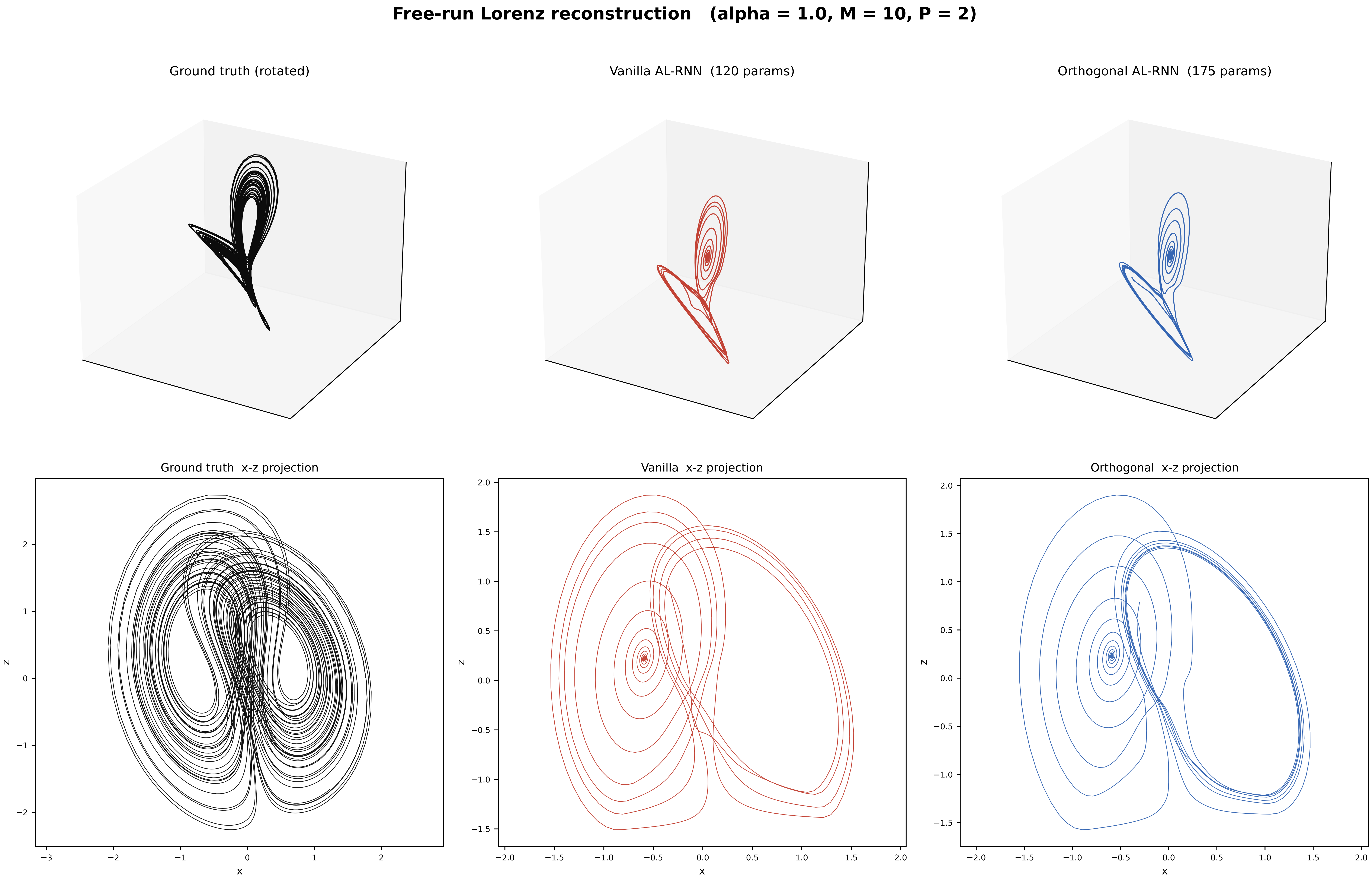


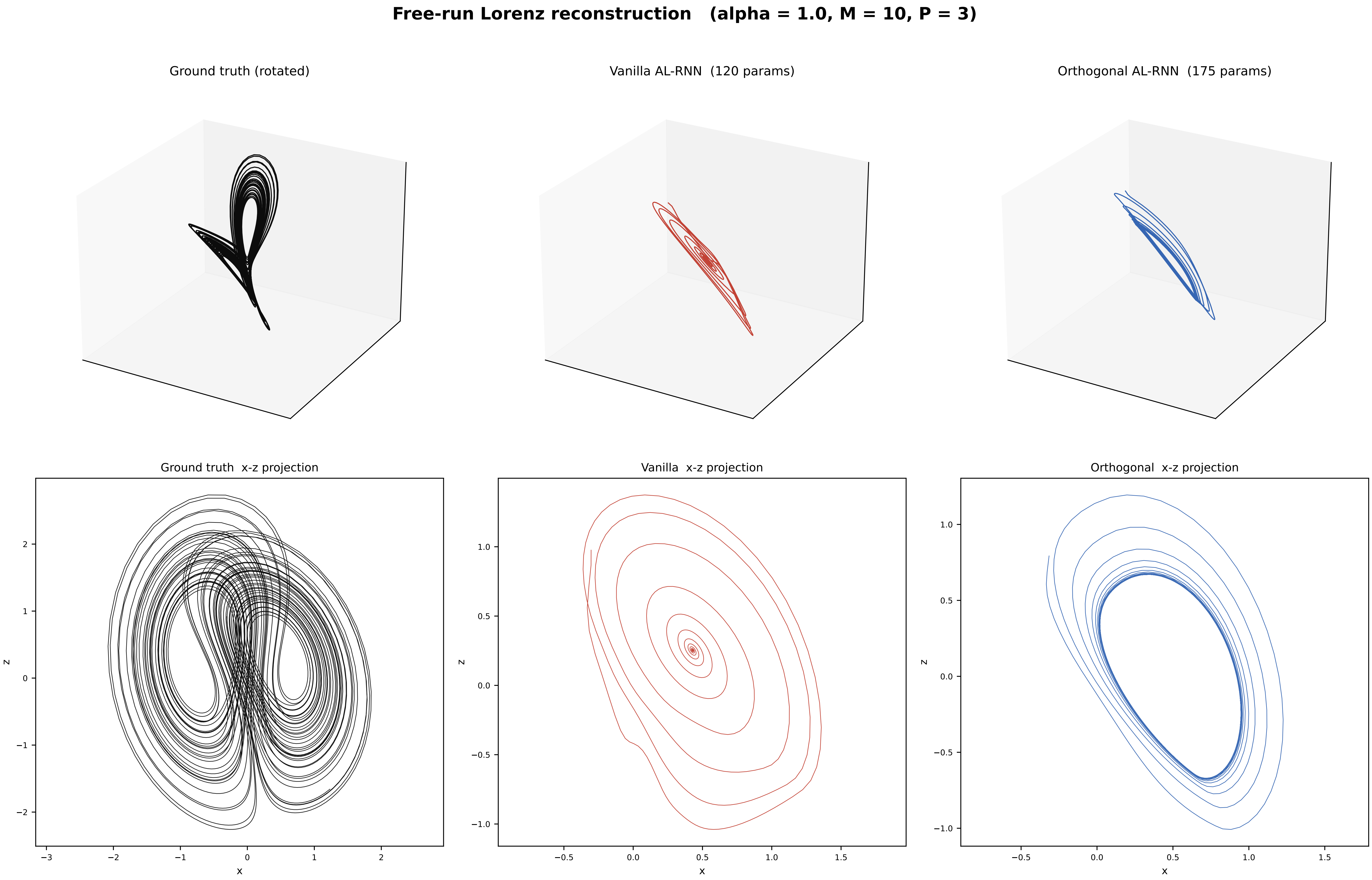
Free-run Lorenz reconstruction (alpha = 1.0, M = 5, P = 2)
Ground truth (rotated)
Vanilla AL-RNN (35 params)
Orthogonal AL-RNN (50 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z



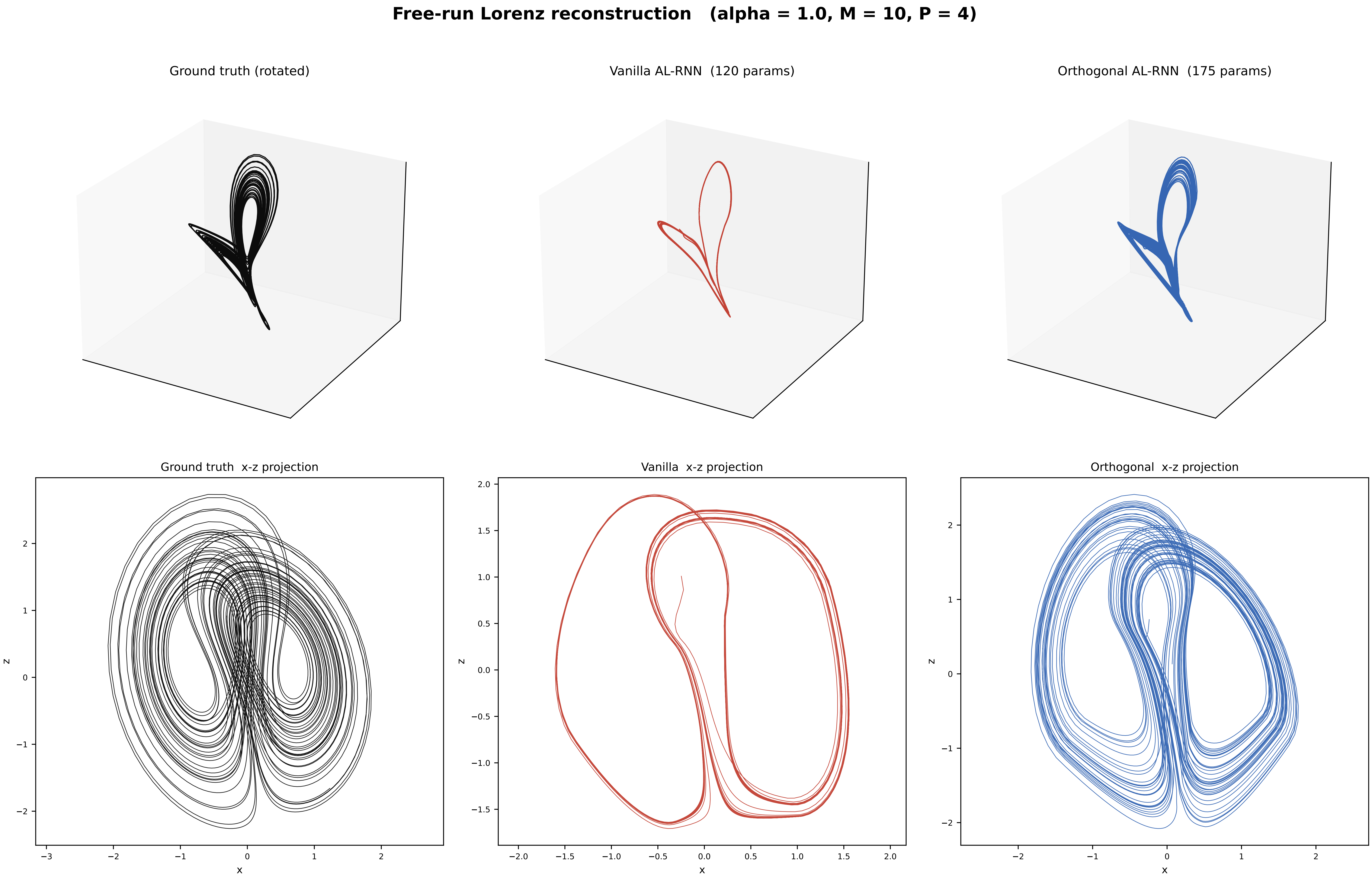
Free-run Lorenz reconstruction (alpha = 1.0, M = 5, P = 3)
Ground truth (rotated)
Vanilla AL-RNN (35 params)
Orthogonal AL-RNN (50 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z

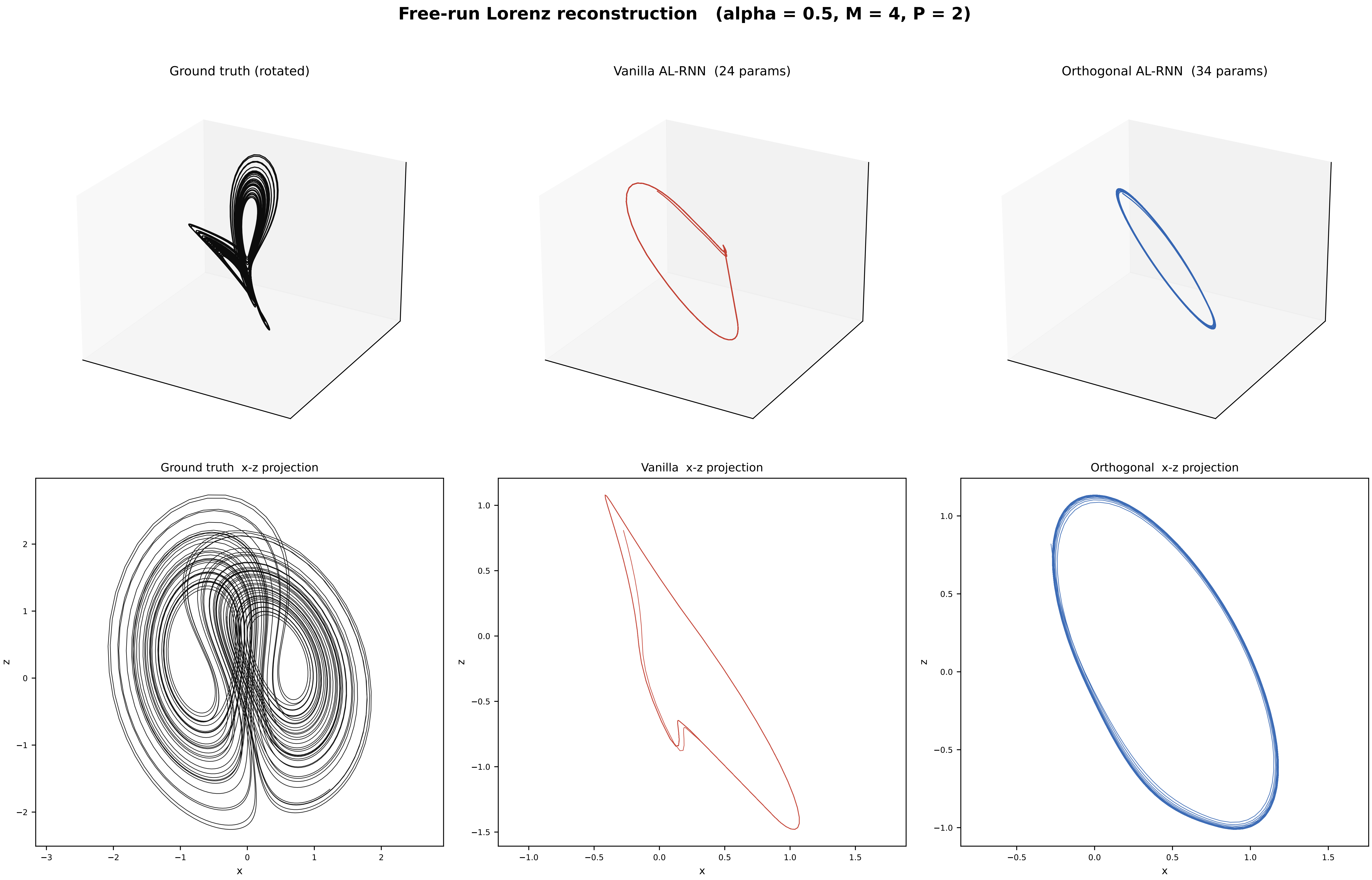

Free-run Lorenz reconstruction (alpha = 1.0, M = 5, P = 4)
Ground truth (rotated)
Vanilla AL-RNN (35 params)
Orthogonal AL-RNN (50 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z


Free-run Lorenz reconstruction (alpha = 1.0, M = 10, P = 2)
Ground truth (rotated)
Vanilla AL-RNN (120 params)
Orthogonal AL-RNN (175 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z

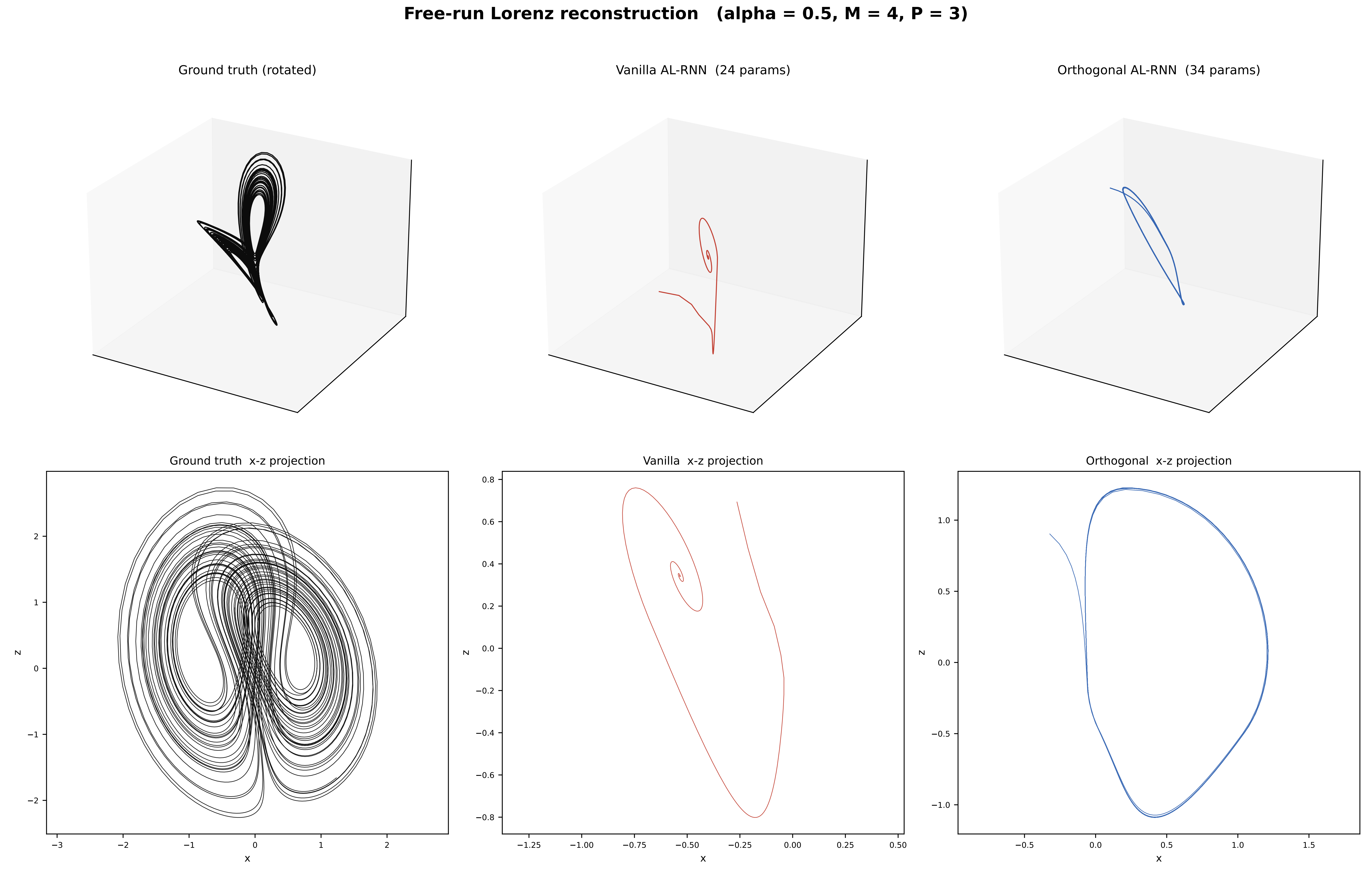


Free-run Lorenz reconstruction (alpha = 1.0, M = 10, P = 3)
Ground truth (rotated)
Vanilla AL-RNN (120 params)
Orthogonal AL-RNN (175 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z


Free-run Lorenz reconstruction (alpha = 1.0, M = 10, P = 4)
Ground truth (rotated)
Vanilla AL-RNN (120 params)
Orthogonal AL-RNN (175 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z

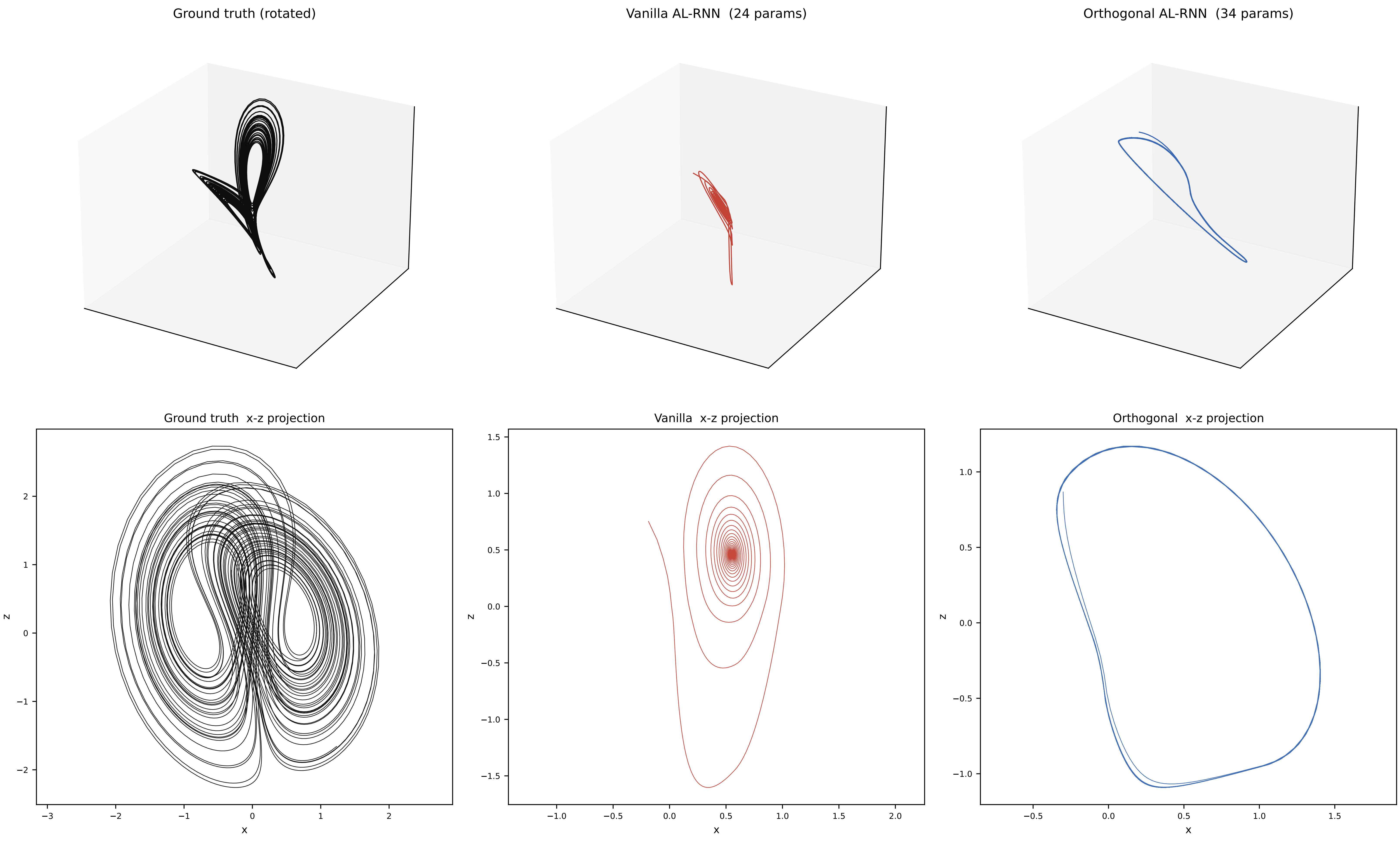


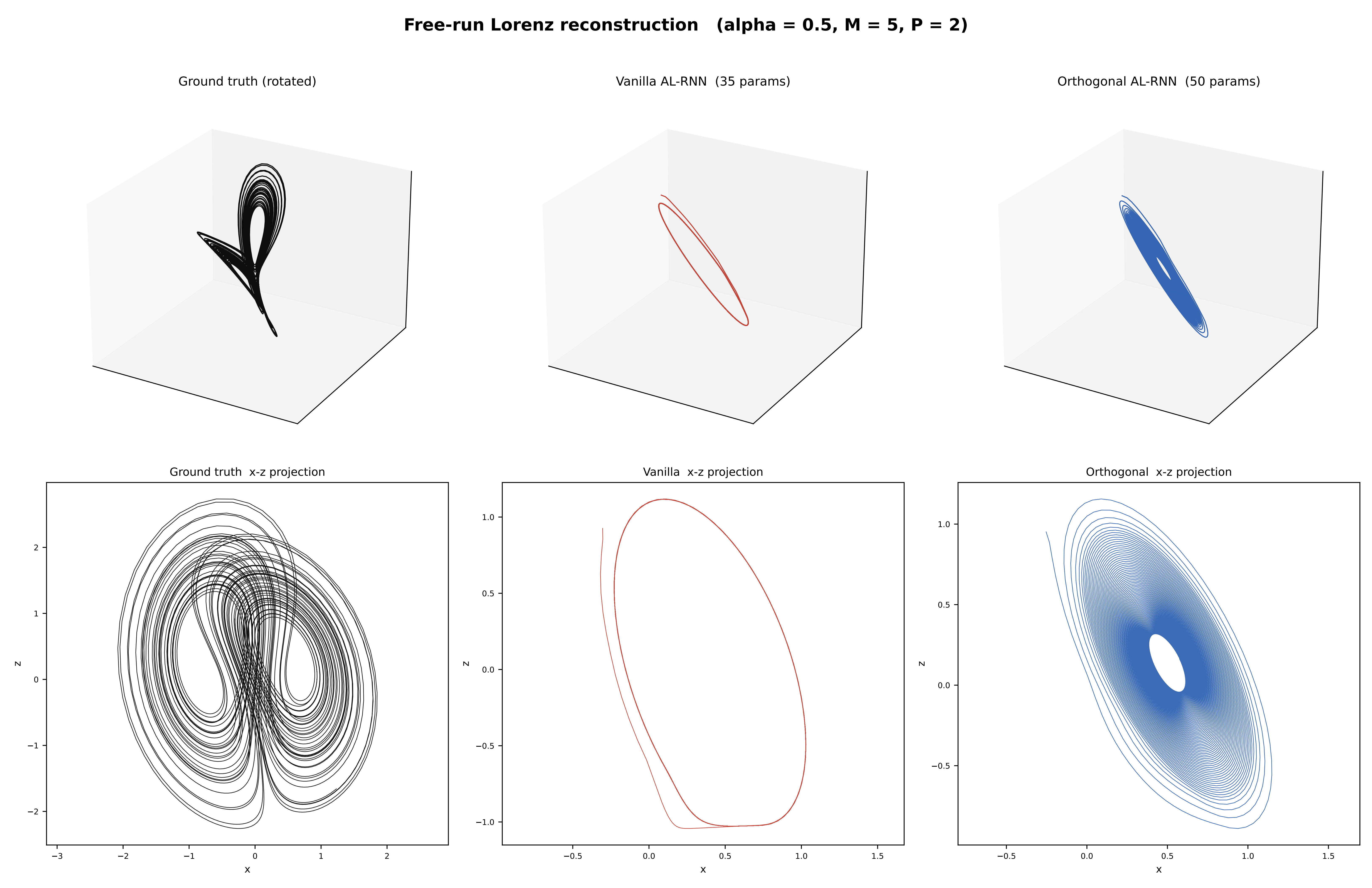
Free-run Lorenz reconstruction   (alpha = 0.5, M = 4, P = 2)
Ground truth (rotated)
Vanilla AL-RNN (24 params)
Orthogonal AL-RNN (34 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z

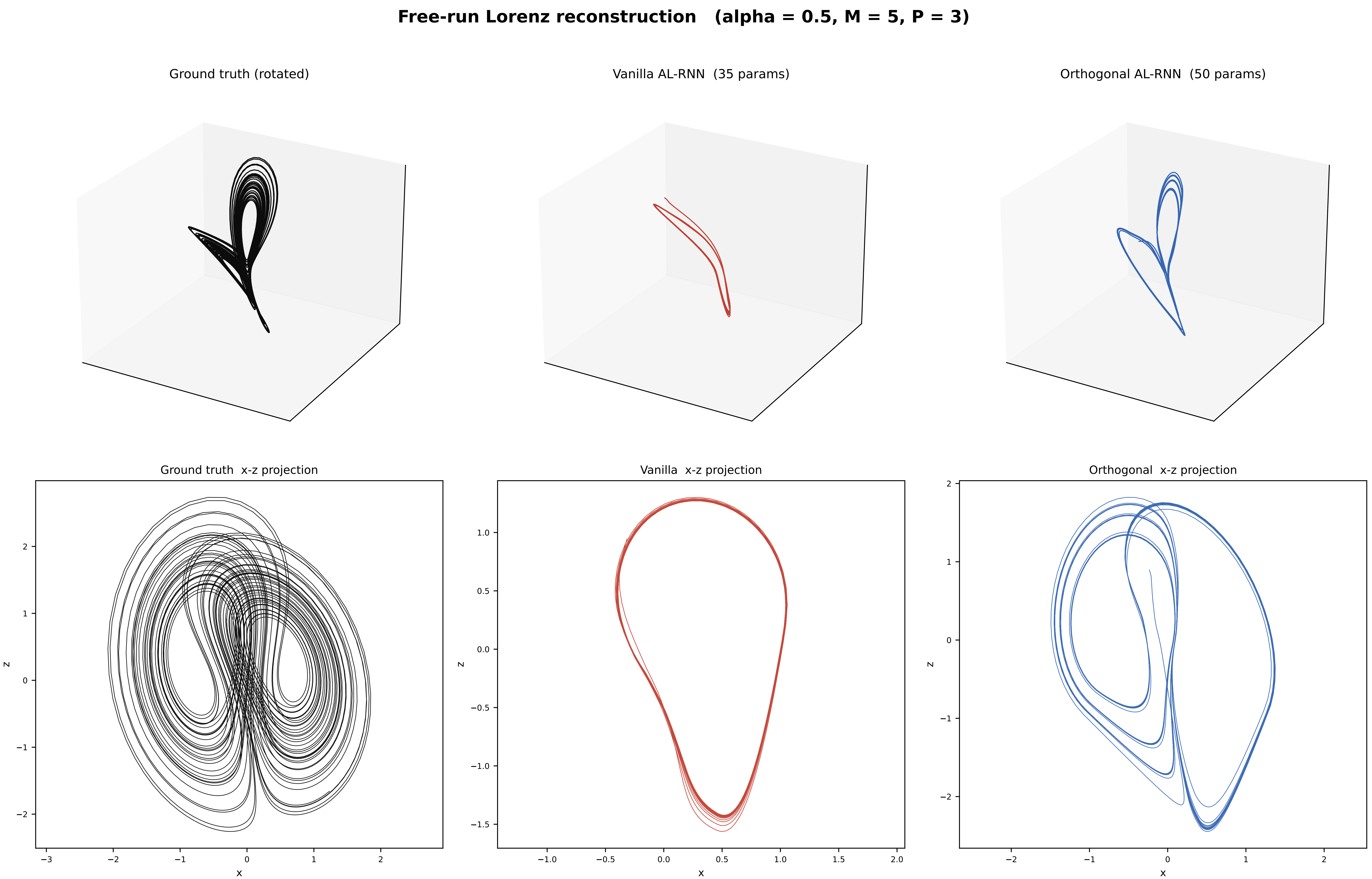


Free-run Lorenz reconstruction (alpha = 0.5, M = 4, P = 3)
Ground truth (rotated)
Vanilla AL-RNN (24 params)
Orthogonal AL-RNN (34 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z

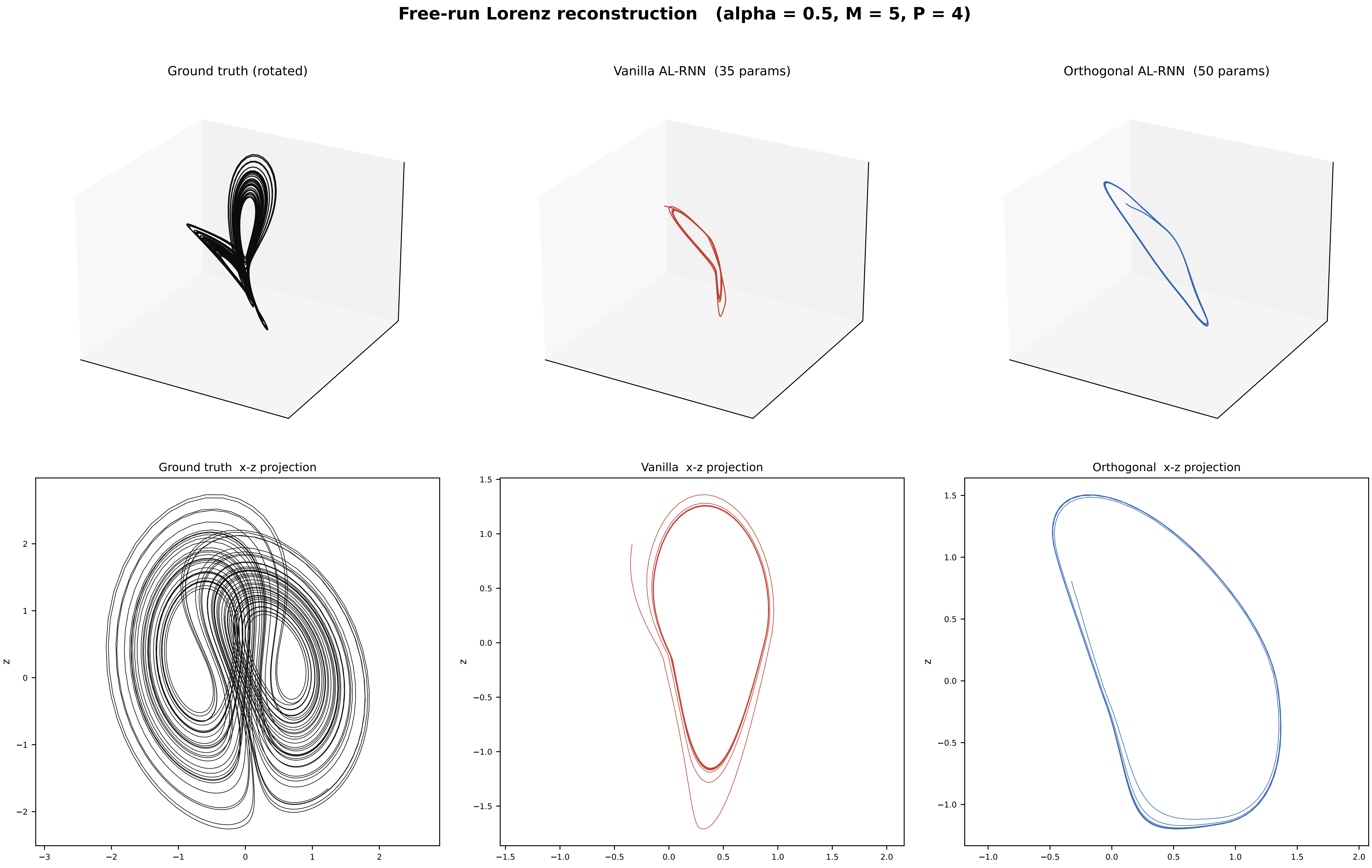
Free-run Lorenz reconstruction (alpha = 0.5, M = 4, P = 4)



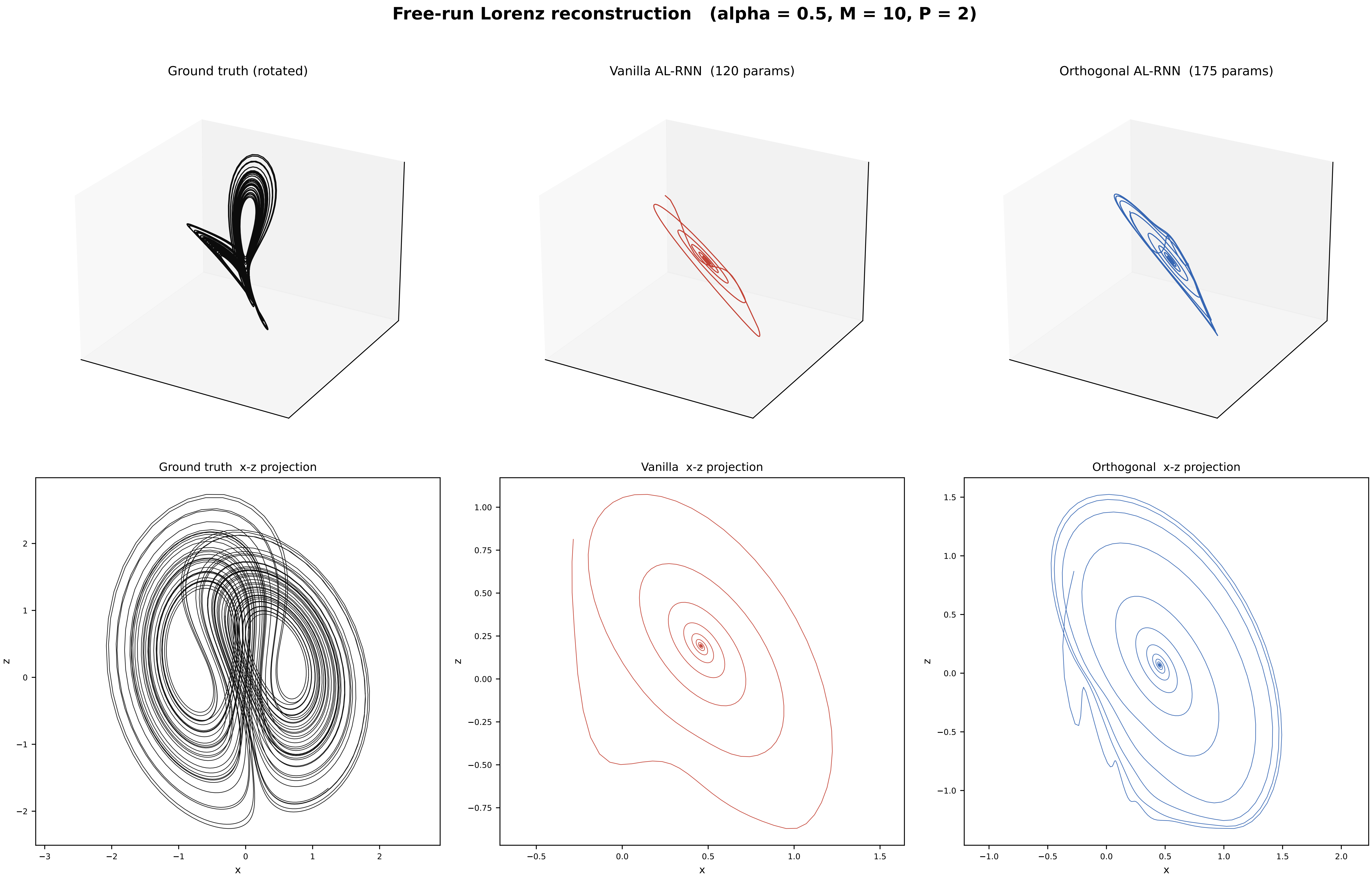

Free-run Lorenz reconstruction (alpha = 0.5, M = 5, P = 2)
Ground truth (rotated)
Vanilla AL-RNN (35 params)
Orthogonal AL-RNN (50 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z

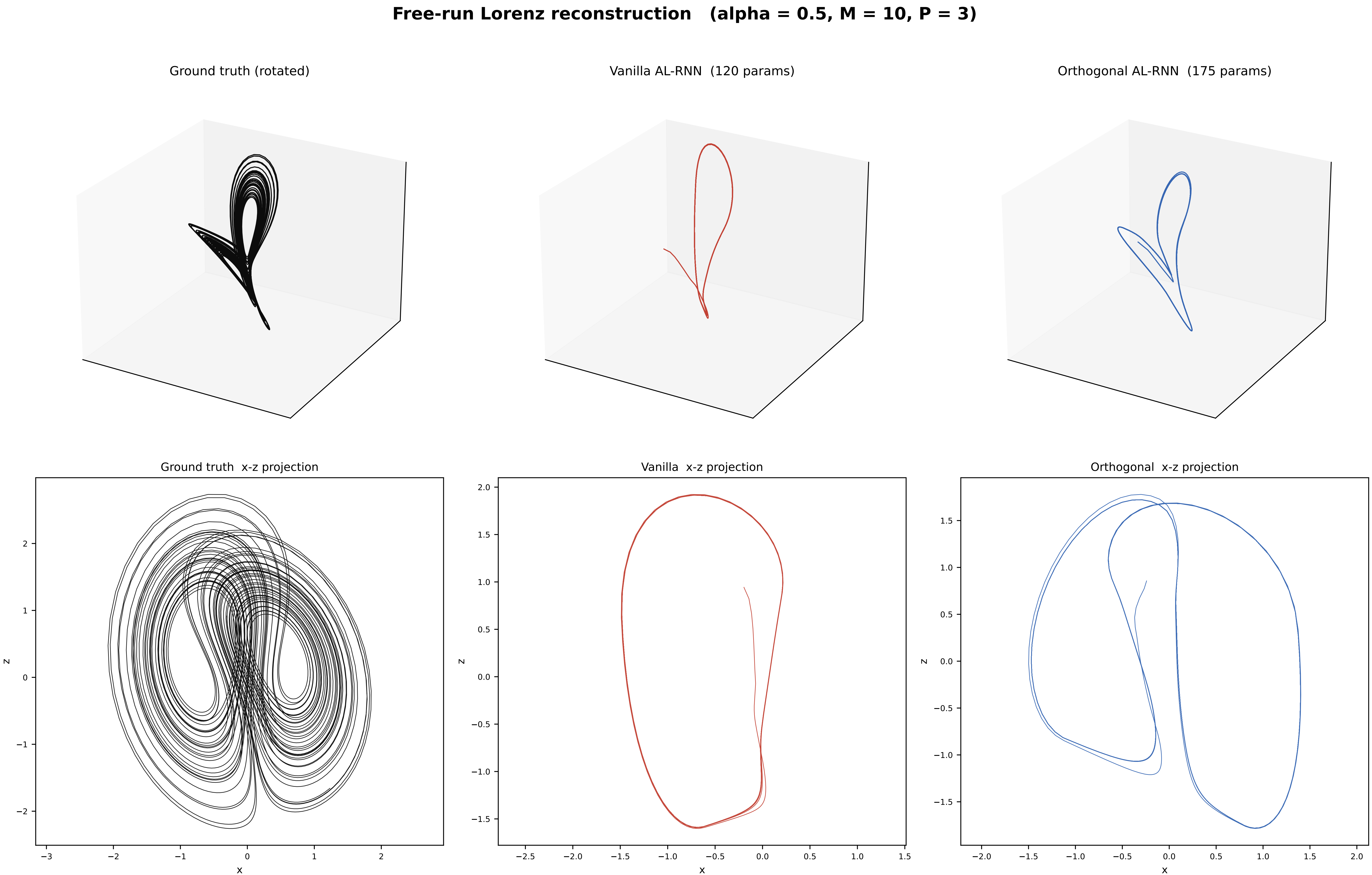


Free-run Lorenz reconstruction (alpha = 0.5, M = 5, P = 3)
Ground truth (rotated)
Vanilla AL-RNN (35 params)
Orthogonal AL-RNN (50 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z



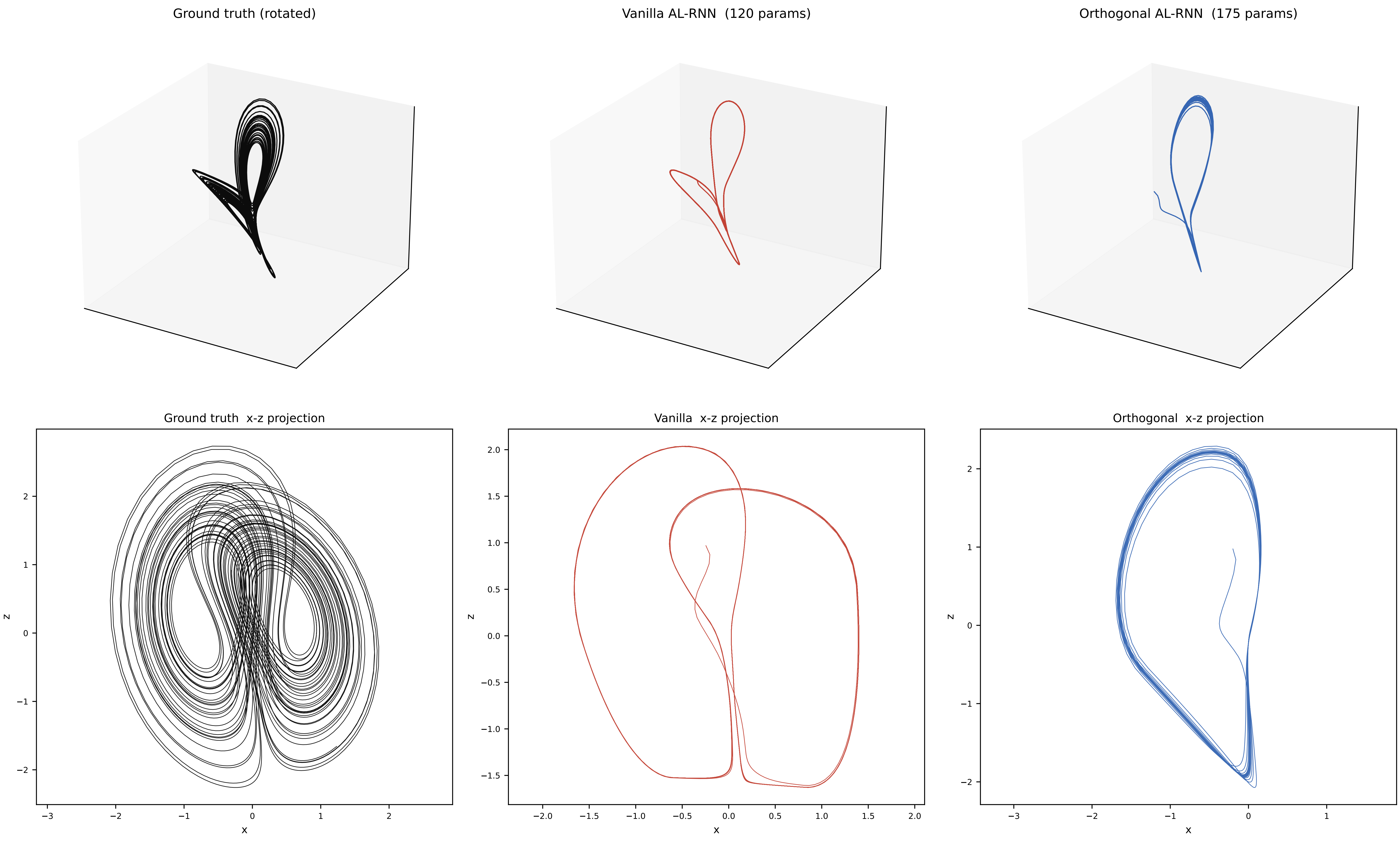
Free-run Lorenz reconstruction (alpha = 0.5, M = 5, P = 4)
Ground truth (rotated)
Vanilla AL-RNN (35 params)
Orthogonal AL-RNN (50 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z


Free-run Lorenz reconstruction (alpha = 0.5, M = 10, P = 2)
Ground truth (rotated)
Vanilla AL-RNN (120 params)
Orthogonal AL-RNN (175 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z

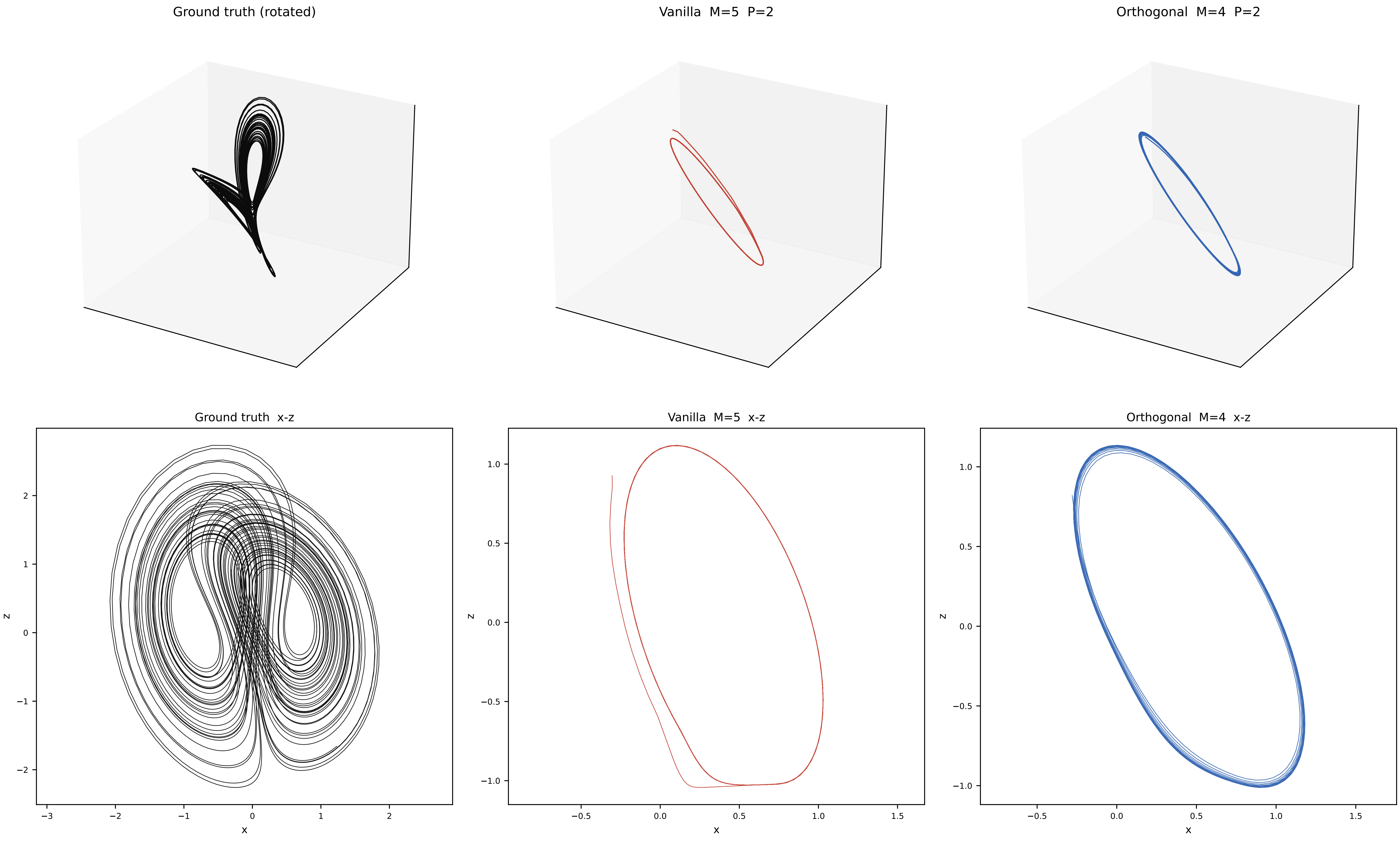


Free-run Lorenz reconstruction (alpha = 0.5, M = 10, P = 3)
Ground truth (rotated)
Vanilla AL-RNN (120 params)
Orthogonal AL-RNN (175 params)
Ground truth x-z projection
Vanilla x-z projection
Orthogonal x-z projection
x
z

Free-run Lorenz reconstruction   (alpha = 0.5, M = 10, P = 4)



Matched pair (alpha = 0.5, P = 2): ortho M=4 (34 params, MSE 0.0663) ≈ vanilla M=5 (35 params, MSE 0.0777) [vanilla uses 1.03× params]



Matched pair (alpha = 0.5, P = 3): ortho M=4 (34 params, MSE 0.0716) ≈ vanilla M=5 (35 params, MSE 0.0691) [vanilla uses 1.03× params]